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AQA · LEVEL 2 · MATHEMATICS

AQA Functional Skills Mathematics Level 2

June 2017 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: AQA-4368-QP-JUN17

100% Built from Empty Question Paper + Official Mark Scheme

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HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Level 2

Data Book (Examination)

Insert

Instructions

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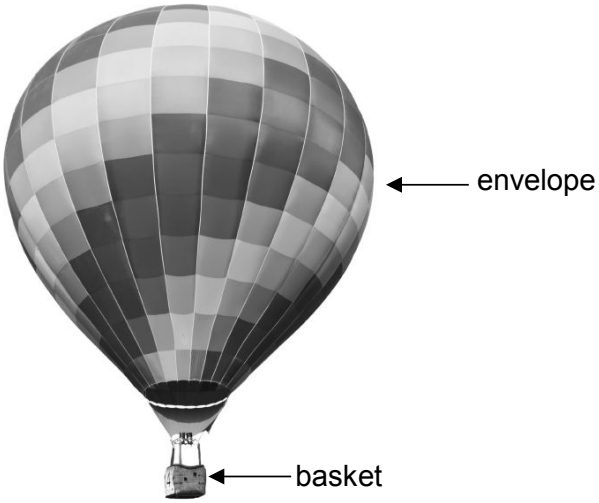
Advice

- This book will not be collected in for marking. Ensure that all working that you wish to have marked is written in the space provided in the question/ answer book.

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Data sheet for Hot Air Balloon

A hot air balloon has an envelope and a basket.



The basket holds the pilot and passengers.

Pilot qualification

Private balloon pilots can qualify to carry passengers.
To qualify, they must

- complete 10 flights
- complete 35 hours of flying
- be the pilot of a balloon that safely rises to 5000 feet
- pass written and practical tests.

Envelope volume

The volume of the envelope is measured in m^3
An envelope with a volume of 2800 m^3 can lift a basket holding up to 5 people.

Height

Use this formula to convert height in metres to height in feet.

$$F = \frac{10M}{3}$$

F is the height in feet
M is the height in metres

Data sheet for Room Makeover

Heat output of a radiator

The heat output of a radiator is measured in British Thermal Units per hour (BTU per hour).
Different types of room need radiators with different heat outputs.

The table shows the room factor (f) for different types of room.

Room type	**Room factor** (f)
Living, dining, bathroom	177
Bedroom	141
Kitchen, stairs, hallway	106

A living room has a higher room factor than a kitchen because it needs more heat.

Use this formula to work out the **minimum** heat output needed for a room.

$$h = v \times f$$

h is the heat output in BTU per hour
v is the volume of the room in m^3
f is the room factor

Radiator size

This table shows some single radiator sizes and some double radiator sizes.
The heat output of the radiators is also shown.

	Heat output (BTU per hour)	
Length (mm)	**Single radiator**	**Double radiator**
500	1768	3316
600	2122	3979
700	2475	4642
800	2829	5305
900	3182	5968
1000	3536	6631
1100	3889	7295
1200	4243	7958



Single radiator



Double radiator

Example

A kitchen has length 4.1 m, width 3.6 m and height 2.7 m

$v = 4.1 \times 3.6 \times 2.7$

$= 39.852$

$f = 106$

$h = 39.852 \times 106$

$= 4224.312$

The kitchen needs a minimum of 4224.312 BTU per hour.

The smallest **single** radiator the kitchen needs has length 1200 mm

The smallest **double** radiator the kitchen needs has length 700 mm

Laminate Flooring



Floors can be covered with laminate flooring.



To lay laminate flooring,

cover the floor with underlay



lay laminate floor planks on top of the underlay



fit edging strips around all the edges of the room.

You can cut these items to fit.

END OF DATA

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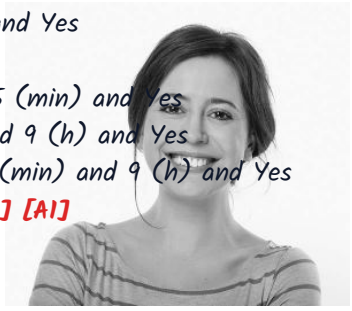
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Answer **all** questions in the spaces provided.

1 **Hot Air Balloon**

There is a **data sheet** for Hot Air Balloon.

Q1(a) Worked Solution:
35 (h) and Yes
12
35 (h) 5 (min) and Yes
9 (h) and 9 (h) and Yes
9 (h) 5 (min) and 9 (h) and Yes
[A2] [A1] [A1]



Jo

Jo is a private balloon pilot.

She wants to qualify to carry passengers.

1 (a) Jo has already completed seven flights.

These flights lasted a total of 26 hours.

She completes three more flights.

	Day	Start time	Finish time
Flight 1	Monday	8.00 am	11.30 am
Flight 2	Friday	10.45 am	1.00 pm
Flight 3	Saturday	2.40 pm	6.00 pm

Jo says,

"I have now completed the hours I need to qualify."

Is Jo correct?

You **must** show your working.

[4 marks]

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1 (b) On one of the flights, the balloon reaches a height of 1546.5 **metres**.

Circle the height that the balloon reaches in **feet**.

[1 mark]

463.95 515.5 5103.45 5155

†1 (c) Jo rents a balloon with an envelope volume of 3198 m^3
A balloon can lift 1 kilogram for every 3.9 m^3 of envelope volume.

Work out the maximum number of kilograms this balloon can lift.

[2 marks]

Check your answer by rounding

3198 to the nearest 100

and

3.9 to the nearest whole number.

[1 mark]

Do not write outside the box

1 (d) Jo rents the balloon for a year.
She makes these notes about the flights and costs for the year.

> There will be
> 7 flights each month in April, May, September and October
> 9 flights each month in June, July and August.
>
> The balloon can hold the pilot and 4 passengers.
> I will charge each passenger £135 for a ticket.
>
> My costs will be
> £5700 for rent and insurance for the year
> £42 for fuel for **each** flight.

Jo says,

"If I sell 80% of the tickets I will make a profit of **more than** £16 000"

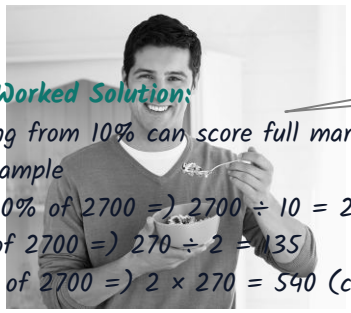
Is she correct?

You **must** show your working.

[7 marks]

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2 **Muesli**



Rory

Rory finds the number of calories in different brands of muesli.

Muesli	**Calories per serving**
Brand W	222
Brand X	170
Brand Y	219
Brand Z	188

He also finds this information.

	Calories
One serving of yoghurt	50
Glass of apple juice	86
Glass of cranberry juice	48
Cup of coffee	25
Cup of tea	10

Question 2(a) · Continued

MFS Stepped Model Solution Continuation Sheet (Overflow-Safe Layout)

(15% of 2700 =) 270 + 135 = 405 (calories)

(b) 270 ÷ 2 = 135

2 × 270 = 540 (calories)

270 + 135 = 405 (calories)

(c) 270 + 135 = 405

270 + 270 = 540

(d) 1% = 27

[M1] [M1]

Do not write outside the box

For a healthy diet, Rory is allowed 2700 calories a day.
He wants his breakfast to have between 15% and 20% of 2700 calories.

2 (a) Show that Rory's breakfast should have between 405 and 540 calories.

[2 marks]

2 (b) For breakfast, Rory has

- two servings of muesli
- two servings of yoghurt
- a glass of apple juice or cranberry juice
- a cup of tea or coffee.

Plan a breakfast that Rory can have.

Include the total calories.

[3 marks]

Do not write outside the box

2 (c) Lara makes her own muesli.
To make 500 g of muesli she needs

325 g oats	20 g dried apricots
35 g nuts	40 g dried cranberries
40 g raisins	40 g sunflower seeds

Lara makes these notes.

> 1 kg of oats costs 80p
> 100 g of nuts cost £1.40
> For 500 g of muesli, the other ingredients cost a total of 96p
> A box of similar muesli from a shop costs £2.94 for 500 g

She says,

"For 500 g, my muesli is cheaper than the shop muesli by **more than** £1.20"

Is she correct?

You **must** show your working.

[6 marks]

Do not write outside the box

2 (d) Sam also makes his own muesli from oats, fruit and seeds.

By weight, he wants

- $\frac{3}{5}$ of the muesli to be oats
- fruit to be four times as much as seeds.

He has 900 g of oats.

He uses all of the oats to make some muesli.

How many grams of fruit does he need?

[4 marks]

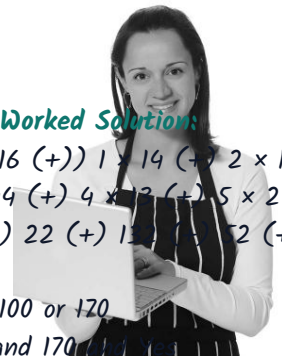
15

Turn over for the next question

Do not write outside the box

3 **Canteen Survey**

Betty is the manager of the college canteen.



I want to improve the service we offer.
To help me I will carry out a survey.

Q3(a) Worked Solutions:
(0 × 16 (+)) 1 × 14 (+) 2 × 11 (+)
3 × 44 (+) 4 × 13 (+) 5 × 2
14 (+) 22 (+) 13 (+) 2 (+) 10
230
1.7 × 100 or 170
230 and 170 [illegible]

Betty

100 students completed Betty's survey form.

Here is one of the completed forms.

[M1] [M1] [A2] [A1]

Canteen Survey

A Circle the number of days that you used the canteen last week.

0 1 (2) 3 4 5

B Tick a box to show which of these sandwiches you like best.

Cheese and Pickle	
Egg and Cress	✓
Bacon, Lettuce and Tomato	
Chicken Salad	
Tuna and Sweetcorn	

C The canteen is thinking of selling sushi.

Tick a box to show what type of sushi you would like best.

Salmon	
Prawn	
Tuna	✓
Vegetable	
None	

Do not write outside the box

3 (a) The mean number of days the 100 students used the canteen that week was 1.7

Betty put some more seats in the canteen.

She then asked the same 100 students question **A** again.

Here is some information about their answers.

Number of days	Number of students
0	16
1	14
2	11
3	44
4	13
5	2
	Total = 100

Betty says,

"The mean number of days the students used the canteen has increased."

Is she correct?

You **must** show your working.

[4 marks]

Do not write outside the box

The table shows information about the answers to question **B**.
It also shows the price of each type of sandwich.

Sandwich	**Number of students**	**Price**
Cheese and Pickle	8	£1.40
Egg and Cress	10	£1.60
Bacon, Lettuce and Tomato	32	£3.00
Chicken Salad	38	£2.00
Tuna and Sweetcorn	12	£3.00
	Total = 100	

Betty usually makes 15 of each type of sandwich each day.

3 (b) Betty decides to double the **total** number of sandwiches she makes.

Show that she will now make 150 sandwiches each day.

[1 mark]

Do not write outside the box

3 (c) Betty uses the information in the table about the answers to question **B**.
She works out how many of each type of sandwich to make.
She then works out how much money she could take from selling the 150 sandwiches.

She says,

"If I sell all the sandwiches I will take **more than** £350 each day."

Is she correct?

You **must** show your working.

[6 marks]

Do not write outside the box

3 (d) Here is some information about the answers to question **C**.

Type of sushi	Tally
Salmon	𝍸 𝍸 𝍸 ||
Prawn	𝍸 𝍸 𝍸 𝍸 𝍸
Tuna	𝍸 𝍸 𝍸 ||
Vegetable	𝍸 𝍸
None	𝍸 𝍸 𝍸 𝍸 𝍸 𝍸 |

Betty wants to show this information in a diagram.

Draw an appropriate diagram that Betty could use.
Use the space below **or** the grid opposite.

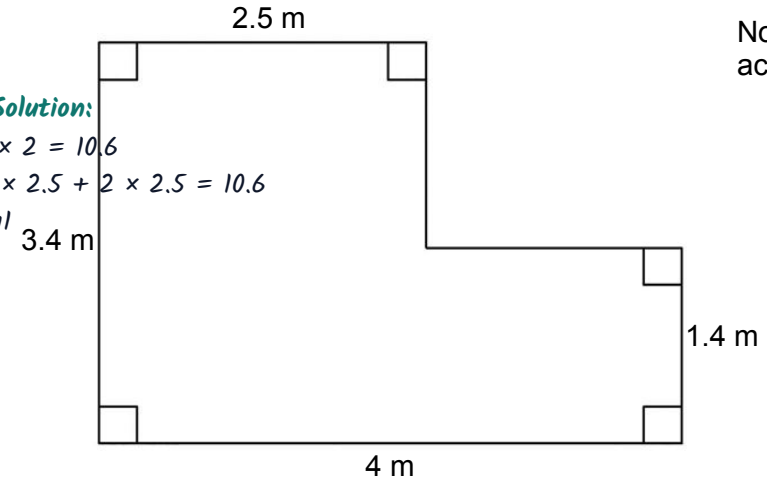
[4 marks]

Do not write outside the box

4 **Room Makeover**

There is a **data sheet** for Room Makeover.

Here is a sketch of Tom's bedroom floor.



Not drawn accurately

Q4(a) Worked Solution:
4 × 3.4 □ 1.5 × 2 = 10.6
1.5 × 1.4 + 1.4 × 2.5 + 2 × 2.5 = 10.6
10.6 × 2.4 × 141
25.44 × 141
3587(.04)
[M1] [A1]

4 (a) Show that the floor area of this bedroom is 10.6 m^2

[2 marks]

4 (b) The bedroom has height 2.4 m
Tom wants a new **double** radiator.

Work out the smallest length of radiator he should buy.

You **must** show your working.

[3 marks]

Do not write outside the box

4 (c) Tom wants to replace the carpet in his bedroom with laminate flooring.
He finds this information.

Foam underlay for laminate flooring

Sold in rolls
£17.95 a roll (covers 15 m^2)

Oak laminate flooring

Sold in packs
£23.80 per pack (covers 1.72 m^2)

Edging

Sold in strips
£1.65 per 2-metre strip

Tom says,

"It will cost under £200 to buy enough underlay, laminate flooring and edging."

Is Tom correct?

You **must** show your working.

[7 marks]

Do not write outside the box

†4 (d) Tom hires Ali to lay the floor.

Ali charges £18 per hour.

He works for $7\frac{3}{4}$ hours.

How much does he charge Tom for $7\frac{3}{4}$ hours work?

[2 marks]

Check your answer.

Show how you have done your check.

[1 mark]

15

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ANSWER IN THE SPACES PROVIDED

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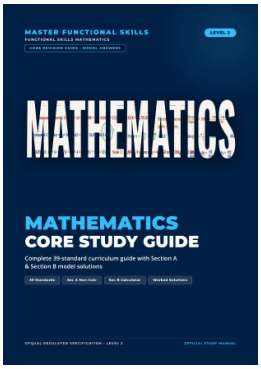
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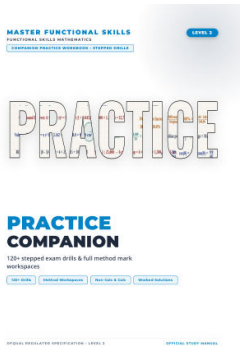


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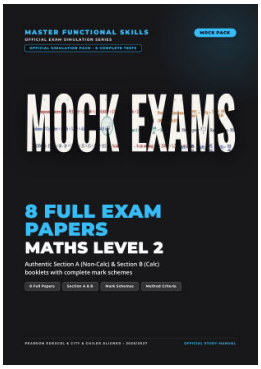


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